

#### Topic 40: Forces – More on gravity

| Students should: |                                                                                                                                        | Guidance<br><i>iLowerSecondary Curriculum reference</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 40.1             | know that the greater the mass of an object, the stronger the gravitational force it exerts                                            | <p>know that the Earth's gravitational force attracts any mass towards its centre</p> <p>know that the force of gravity on a given object is less on the Moon than on the Earth</p> <p>appreciate that a given object will weigh less on the Moon than on Earth</p> <p><i>Year 8: Forces – More on gravity</i></p>                                                                                                                                                                                                                             |
| 40.2             | know that the approximate value of the gravitational field strength (g) on the surface of Earth = 10 N/kg                              | <p>know that gravitational field strength (g) is smaller on the Moon's surface than on the Earth's surface</p> <p><i>(This builds on Year 7: Forces – Different types of forces)</i></p> <p><i>Year 8: Forces – More on gravity</i></p>                                                                                                                                                                                                                                                                                                        |
| 40.3             | know how to use the formula:<br>weight = mass (m) x gravitational field strength (g)                                                   | <i>Year 8: Forces – More on gravity</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 40.4             | know that the gravitational force of Earth acting on an object decreases as the object moves further away from the centre of the Earth | <i>Year 8: Forces – More on gravity</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 40.5             | know that gravitational force causes moons to orbit planets and causes the planets to orbit the Sun                                    | <i>Year 8: Forces – More on gravity</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 40.6             | know that gravitational force causes artificial satellites to orbit the Earth and causes comets to orbit the Sun                       | <p>know the terms <i>natural satellite</i> and <i>artificial satellite</i></p> <p>know that satellites can be used to orbit the Earth and other planets to take photographs and provide scientific data, for example temperatures</p> <p>know some uses of artificial satellites for example:</p> <ul style="list-style-type: none"> <li>• polar orbiting satellites for observation, weather forecasting and reconnaissance</li> <li>• geostationary satellites for communications.</li> </ul> <p><i>Year 8: Forces – More on gravity</i></p> |